

# ARC

When electricity jumps between two conducting electrodes, such as two exposed pieces of wire, it makes an **electric arc**. The air between the electrodes becomes ionized, allowing electron to jump across and create a visual **bridge**.

An electrical arc can produce high heat and power - full light, making it useful in areas such as **welding**.

**Lightning** is an example of a naturally occurring electric arc the atmosphere is ionized and creates a visual spectacle.

# WORKING

An electric arc is a device in which an electric

current is caused to flow (a flow of electron) between two points separated by a gas. These two points are called **electrodes**.

The one from which the current originates is the **cathode**.

The electrode toward which electrons flow is the **anode**.

The term electric arc refers both to the **DEVICE** itself as well as to the **ELECTRIC DISCHARGE** that takes place within the device.

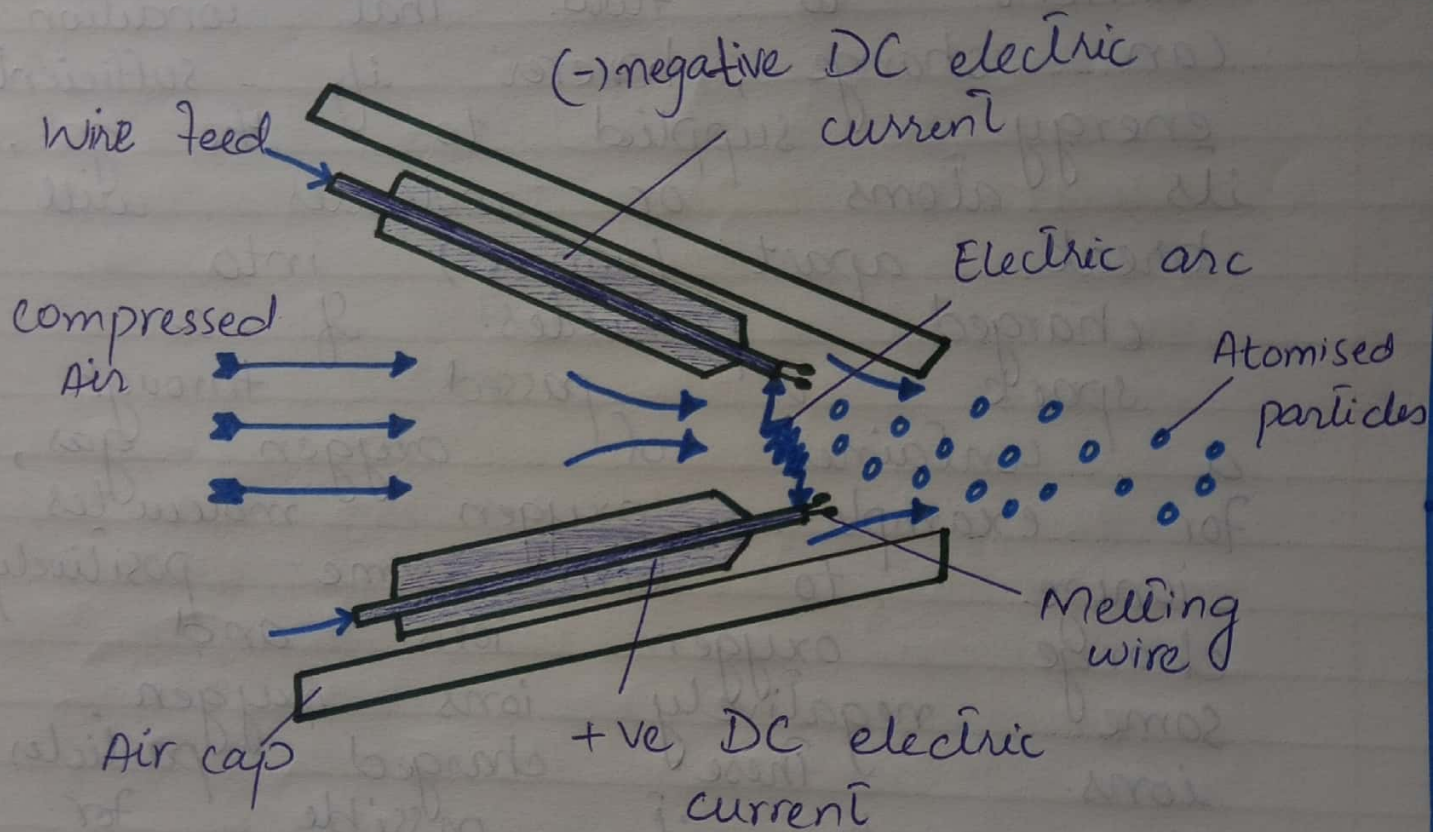
Arc can make use of high, atmospheric, or low pressures and can contain a variety of gases.

## **ELECTRICAL CONDUCTIVITY IN GASES**

Gases are normally poor



conductors of electricity. The atoms or molecules of which they consist usually contain no free electrons needed for a current to flow. That condition can change, however if sufficient energy is supplied to the gas, its atoms or molecules will break apart (ionize) into charged particles. If a spark is passed through a container of oxygen gas, for example, oxygen molecules ionize to form some positively charged oxygen ions and some negatively charged oxygen ions. These charged particles then make it possible for the gas to become conductive.



**Fig : ELECTRIC ARC  
OR  
ARC DISCHARGE**

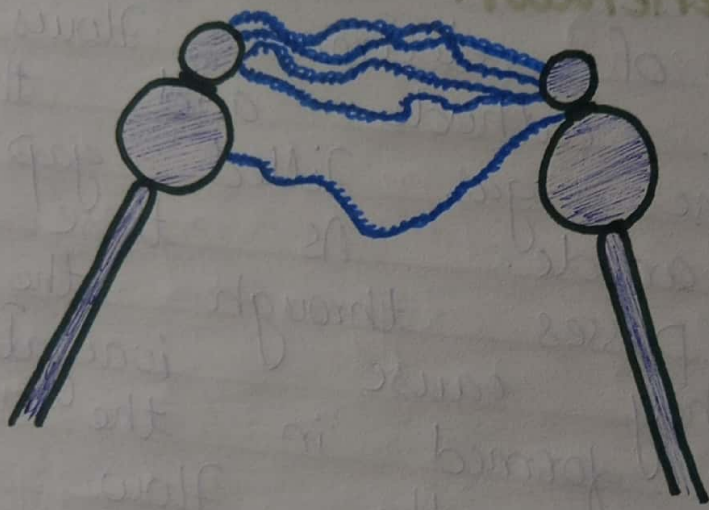


# ARC CONSTRUCTION

In an electric arc, the energy needed to produce ionization comes from an external source, such as an **electric generator**. An intense stream of electrons flows into the cathode and then across the gas filled gap to the anode. As these electrons pass through the gas, they cause ionization. Ions formed in the process make the flow of current between electrodes even easier. For every gas, some minimum amount of energy is needed, to produce ionization at given temperature and pressure. That energy is known as the **GAS'S BREAKDOWN POTENTIAL**.

## EXAMPLE:

One example of electric arc is a **lightning**



**Fig: ELECTRIC ARC**

**POTENTIAL  
GAS'S**

**Example:  
BREAK DOWN**



**bolt.** In nature, two clouds can act as electrodes, or electric current may flow between a cloud and earth's surface. In either case current flows through the air, ionizing molecules of OXYGEN, NITROGEN, and other gases in the atmosphere.

The light and sound associated with lightning are evidence of an important change that occurs in the gas between electrodes. The flow of electric current heats the gas to higher temperature. The light associated with lightning is evidence of that change. The clap of thunder is another example of / sign of the change - the heated air around the lightning bolt expands rapidly, producing a sound wave.

The simplest arc consist of 'Two ELECTRODES' made of a conducting material and situated a short distance from each other.

AIR IS THE GAS USED IN THIS ARC.

This kind of electric arc was first studied by English physicist and chemist **Humphry Davy** (1778-1829) in 1808.

Various types of electric arcs differ from each other in two respects.

- ① The pressure at which they operate.
- ② The materials of which they are made.

Electric arcs may be enclosed in glass or plastic containers from which air has been pumped out (vacuum arcs) or



to which air or some other gas has been added (high pressure arcs).

The light produced by an arc depends both on the material from which the electrodes are made and on the gas that separates them. Some electrodes have no function other than to conduct an electric current, into and out of the arc. Other electrodes are chosen because they tend to be vaporized when the arc is used, changing the discharge that is produced. Various gases are chosen for use in electric arcs because they too affect the discharges produced. For instance, each chemical element produces its own characteristic colour when ionized.

# USES OF ELECTRIC ARCS

Many types of arcs exist, each with its own applications.

## For example **FOR WELDING:**

Arc welders are used for welding (where a metal is fused and added in a joint).

## **FOR HEAT SUPPLY:**

In some cases, the arc's only function is to supply heat.

## **PLASMA TORCH:**

Plasma torches are used for cutting, spraying and gas heating. Plasma is a term used for hot, ionized gases. Cutting a metal with plasma torch may be done by means of an arc formed between the metal itself and an electrode from the torch.

## **ARC LAMPS:**

Electric arcs may be used as lamps because of the amount



of light they produce.

## **CARBON ARCS:**

The carbon arc, in which two carbon rods serve as electrodes, was the first practical commercial lightning device.

Carbon arcs are the brightest source of light and used in theater motion picture projectors, large search-lights and lighthouses.

## **FLAME ARCS:**

Flame arcs are used in colour photography and in photo-chemical processes because they closely approximate sunshine.

Color of flame arcs depends upon the material of which electrodes are made. For example

Calcium arc → Red glow

Barium arc → Green glow

## **RADIO VALVES, IN NUCLEAR REACTORS:**

Arcs can also be used in radio valves, such as those used in early days of radio,

and as a source of ions in nuclear reactors. and thermonuclear devices.

### **FOR PLASMA CUTTING:**

Industrially arcs are used for plasma cutting.

### **FOR ELECTRIC DISCHARGE MACHINING:**

Arc is used for electrical discharge machinery.

### **ELECTRIC ARC FURNACES:**

Electric arc furnaces are used to produce steel and other substance.

### **XENON ARC LAMPS:**

Xenon arc lamps are used for movie projectors.

### **OTHER APPLICATIONS:**

They are also used as rectifiers, switching, protection, fuses -